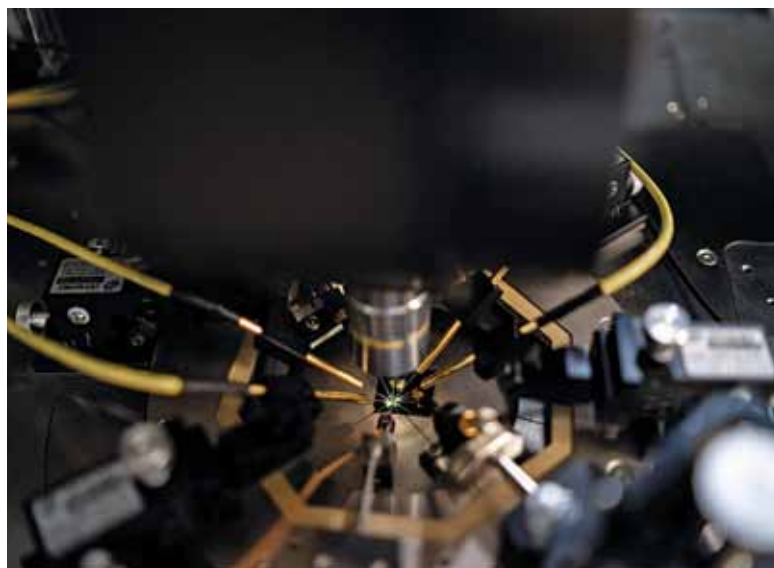




RESEARCH & INNOVATION UPDATES

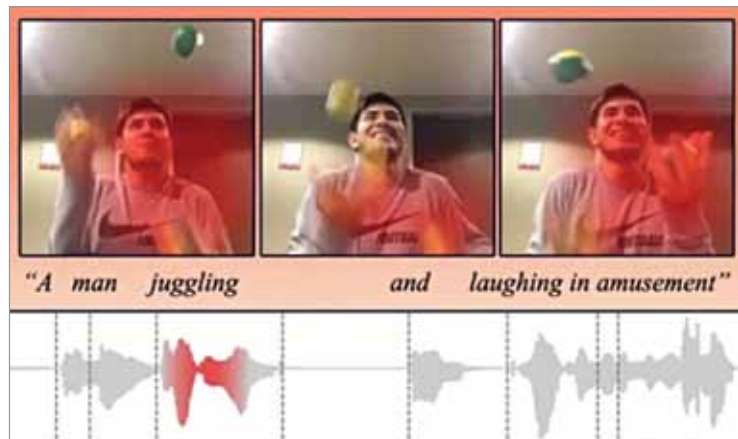
Deep belief neural network made based on silicon memristive synapses

Researchers at Technion-Israel Institute of Technology and the Peng Cheng Laboratory have recently created a new neuromorphic computing system supporting deep belief neural networks (DBNs), a generative and graphical class of deep learning models. This model is based on silicon based memristors, which are energy-efficient devices that can both store and process information. The artificial synapses were fabricated using commercial complementary-metal-oxide-semiconductor (CMOS) processes. These silicon based synapses have numerous advantageous features, including analogue tunability, high endurance, long retention time, predictable cycling degradation, and moderate variability across different devices. The results showed over 97% accurate recognition of handwritten digits when using Y-Flash based memristors.



Memristors measured in a probe station (Source: <https://techxplore.com>)

AI system learns concepts shared across video, audio, and text



MIT researchers developed a machine learning technique that learns to represent data in a way that captures concepts which are shared between visual and audio modalities. Their model can identify where certain action is taking place in a video and label it (Source: <https://scitechdaily.com>)

Visual learning is considered a highly efficient way to understand a concept. Researchers at the Computer Science and Artificial Intelligence Laboratory (CSAIL) have developed an artificial intelligence (AI) technique that allows machines to learn concepts shared between different modalities, such as videos, audio clips, and images. The model learns to represent data in a way that captures concepts which are shared between visual and auditory modalities. The work focuses on representation learning, which is a form of machine learning that seeks to transform input data to make it easier to perform a task like classification or prediction. The model still has some limitations that need to be addressed. For one, their research focused on data from two modalities at a time, but in the real world humans encounter many data modalities simultaneously.